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Applying Machine Learning to Hydrodynamics for the Generalization of Multiple Geometries

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1 Literature Review

1.1 Introduction

Hydrodynamics is one of the diverse fields of study in the current academic landscape. It encompasses a broad range of disciplines, from physics and biology to chemistry and medicine. However, in such a large field, it is impossible to encompass all aspects in one study. Therefore, we will be narrowing down the scope to interactions at low Reynolds (low-Re) numbers. Hydrodynamic interactions at low Reynolds number make up the bulk of the field of complex fluid dynamics (CFD) and colloidal suspensions (Happel and Brenner, 1983). This is further observed when particles move in confined geometries, such as doubly periodic domains or slit channels (Maxian et al., 2021; Peláez, 2022). Foundational work on Stokes flow (Happel and Brenner, 1983) and mobility tensors (Ermak and McCammon, 1978; Yamakawa, 1970), such as the Oseen and Rotne-Prager-Yamakawa models, have been essential to establishing robust mathematical frameworks for the aforementioned systems (Dhont, 1996). On the other hand, modern numerical methods, such as Stokes-based methods, allow simulating environments with thousands of particles in complex domains (Brady and Bossis, 1988; Sierou and Brady, 2001).

Taking into consideration these methods, some studies aim to show that the use of machine learning (ML) for fluid mechanics could be a shift to data-driven and physics-informed models for accelerated computational fluid dynamics with comparable accuracy and significant acceleration (Brunton et al., 2020; Kashinath et al., 2021). For instance, hybrid approaches using convolutional neural networks (CNNs) for CFD have achieved significant speedups for Navier-Stokes solvers (Guo et al., 2016; Kochkov et al., 2021), and recent surveys show a significant use of such methods, such as neural operators, graph neural networks (GNNs) and physics-informed neural networks (PINNs) for a wide range of applications such as flow prediction, closure modeling and flow control (Brunton et al., 2020; Vinuesa and Brunton, 2023). However, most of the current work is based on continuum Navier-Stokes systems at moderate or high Reynolds (high-Re) numbers with simple, unchanging geometries (Duraismy et al.,

2019; Vinuesa and Brunton, 2023). Therefore, this work aims to apply ML methods to low-Re hydrodynamics in many-particle systems, with a special emphasis on the generalization across multiple geometries where hydrodynamic interactions play a central role (Peláez, 2022; Maxian et al., 2021).

Therefore, it is important to note that the scope will be restricted to methods which either learn or accelerate particle-based hydrodynamic solvers. High-Re systems, modelling, generic ML-CFD for fields other than hydrodynamics, and purely experimental applications without a modelling component will only be considered and discussed when providing significant insights, such as to generalization, architecture, or evaluation methodologies.

Method-based approaches

1.2 Physics-based vs machine learning vs hybrid approaches

Classic physics-based approaches for hydrodynamic interaction are usually a combination of low-Re Stokes flow theory, analytical or numerical mobility tensors, and well-defined boundary conditions (Happel and Brenner, 1983; Dhont, 1996). Another approach, which sees relevance and use, is Stokesian Dynamics augmented using spectral Ewald acceleration, especially for many-body hydrodynamics, but it has $O(N^2)$ computational bottlenecks (Brady and Bossis, 1988; Lindbo and Tornberg, 2012; Sierou and Brady, 2001). Other options are Graphical Processing Unit (GPU)-based Brownian Dynamics with Hydrodynamic Interactions (BDHI) frameworks such as Universally Adaptable Multiscale Molecular Dynamics (UAMMD) are able to work around the bottleneck by implementing solvers with Fast Fourier Transform (FFT) based algorithms which are able to reduce the complexity from $O(N^2)$ to $O(N) - O(N \log N)$ while enforcing balance of fluctuation and dissipation, and incompressibility within the system (Peláez, 2022; Maxey and Patel, 2001; Fiore et al., 2017).

On the other hand, ML-based approaches are comprised of data-driven models, PINNs, and hybrid strategies. These approaches are able to learn flow evolution from experimental data, but have shown difficulties in the generalization of geometries and boundary conditions (Guo et al., 2016; Duraisamy et al., 2019). In particular, PINNs are interesting due to the fact that they are able to encode partial differential equations (PDEs) and boundary conditions into their

own loss, which enables them to find both forward and inverse solutions to Navier-Stokes and other similar problems but have issues with complex domains and multi-scale problems (Raissi et al., 2019; Wang et al., 2022; Krishnapriyan et al., 2021).

Finally, hybrid methods are based on joining ML predictions to existing solvers or learning corrections to closure models (Kashinath et al., 2021; Caron et al., 2024). Furthermore, some studies have shown that hybrid models are able to achieve favorable speedup-accuracy tradeoffs and even increased results in Navier-Stokes problems (Kochkov et al., 2021; Guo et al., 2016; Vinuesa and Brunton, 2023).

1.3 Physics-inspired Neural Networks vs Set Transformers vs Graph Neural Networks

One of the most important considerations for how to solve the problem of hydrodynamic interactions is the model choice. There are three models which show significant potential for solving the problem of hydrodynamics and the generalization of multiple geometries, which are Physics-inspired neural networks (PINNs), Set Transformers, and Graph Neural Networks (GNNs).

First, PINNs such as Hamiltonian and Lagrangian neural networks enforce invariance in the system through parameters such as energy or momentum conservation and are able to use knowledge from known symmetries to reduce data requirements (Greydanus et al., 2019; Cranmer et al., 2020). Furthermore, $E(3)$ -equivalent architectures are able to further expand on this idea by outputting vectors such as velocity which are able to more easily transform under different rotations and translations, making them especially useful for low-Re hydrodynamics where translational and rotational invariance play a huge role (Thomas et al., 2018; Batzner et al., 2022). With these models, we are also able to encode other properties such as symmetry of the mobility tensor and invariance under rigid conditions (Thomas et al., 2018; Batzner et al., 2022).

Second, Set Transformers are useful since they treat particle configurations as unordered sets and make use of the attention mechanism (Vaswani et al., 2023) to model particle interactions and maintaining permutation invariance (Lee et al., 2019). This is the case since Set

Transformers build upon the traditional transformer by using the self-attention mechanism to capture long term dependencies like those observed in hydrodynamic interactions, and can even scale from $O(N^2)$ to near-linear complexity by using inducing-point strategies or hierarchical strategies (Lee et al., 2019). For our case, Set Transformers are able to represent every particle as an element of the set with different features to encode position, force, and local geometry parameters, and use the attention to learn different kernels which can be close to mobility vectors (Lee et al., 2019; Vaswani et al., 2023). They are particularly suited for hydrodynamics and generalization for their flexibility and performance on set-structured tasks, but they can be computationally expensive and may need some adaptation by adding a physics-inspired loss (Brunton et al., 2020; Cranmer et al., 2020)

Finally, GNNs offer an interesting representation of the system by explicitly modelling particles as nodes and hydrodynamic relationships as edges (Battaglia et al., 2018; Sanchez-Gonzalez et al., 2020). Because of the information propagation nature of message passing in GNNs along edges, being able to be defined by interaction radii and neighbour lists, they are able to handle variable particle numbers and non-uniform configurations (Sanchez-Gonzalez et al., 2020). Furthermore, advances in tensor-field networks provide specific architectures where both edge and node features are automatically translationally and rotationally transformed, which opens up the learning of vector-valued hydrodynamic representations (Thomas et al., 2018; Batzner et al., 2022). GNNs are also able to allow for sparsity by limiting edge reach to near-neighbours or even hierarchical pooling to improve scalability for larger systems when compared to dense attention (Battaglia et al., 2018; Sanchez-Gonzalez et al., 2020). The main issue with these algorithms is that they struggle with long-range interactions, which may need specialized message-passing schemes or deeper networks, and they also struggle with maintaining global properties over the whole system (Sanchez-Gonzalez et al., 2020; Batzner et al., 2022).

1.4 By Application Domain

The application of hydrodynamics can change drastically depending on the domain and how the system is arranged. For instance, quasi-two-dimensional and confined geometries are used

to describe biological systems, and theoretical work has shown that depending on how the confinements are defined, it can change how the mobility tensor diverges, which can lead to anomalous collective diffusion (Panzuela et al., 2017; Peláez et al., 2018). In this case, GPU-based algorithms such as UAMMD can be used to apply BDHI to these geometries, which shows the possibility of extending ML algorithms across geometries (Peláez, 2022).

Another example of how geometry generalization can be used is by looking at spectral methods developed for electrostatics in doubly periodic slit channels, which are Poisson problems with complex boundary conditions (Peláez et al., 2018). This highlights how Fourier-based approaches can be generalized to confined geometries, which, when combined with ML algorithms, can be useful for studying geometries and material properties.

1.5 Identified Gaps and Motivation

Even with the extensive literature and academic works on ML-CFD and the readily available GPU-accelerated BDHI algorithms, there is no specific research on the combination of ML to generalize across multiple geometries. For instance, Deep Learning studies on Reynolds-Averaged Navier-Stokes simulations, which are explicitly trained on a fixed set of geometries, are unable to generalize well enough to other unseen geometries to justify their use, especially when considering new shapes and boundary conditions (Ling et al., 2016; Duraisamy et al., 2019). Furthermore, existing benchmarks are focused mainly on continuum Navier-Stokes problems applied to simple domains, which leaves other, more complex geometries untested and unable to be effectively worked on (Zhang et al., 2023; Vinuesa and Brunton, 2023).

On the other hand, UAMMD and other solvers are able to generate high-quality data across more complex geometries, proving that these more complex geometries can also be generalized (Peláez, 2022; Panzuela et al., 2017; Maxian et al., 2021). This shows a chance for a ML based model to be able to be developed, with it having to be geometry-aware and feasible to train and deploy. Finally, the motivation is to design a GNN, PINN, and Set Transformer to evaluate the architectures and check if it is possible to develop a robust, generalizable system in complex domains.

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